

# When the Lights Go Out: A Hydrogen-Battery Hybrid Microgrid as a Lifeline for Rural Africa

Thato Maelane

Department of Electrical Engineering

Tshwane University of Technology

Pretoria, Gauteng, South Africa

Thato6216@gmail.com

**Abstract**—In Limpopo Province, South Africa, rural health clinics face a crisis that rarely appears in national headlines: when storm clouds persist for two or three days, solar panels stop producing power, batteries go flat, and the lights go out. Vaccines spoil. Water pumps stop. Emergency theatre lighting fails. This paper asks a direct question: can hydrogen storage prevent that outcome, affordably? A 96-hour computational energy balance simulation, modelling a severe winter overcast anomaly representative of the Limpopo highveld climate, shows that a solar-plus-battery-only system (System A) leaves a rural clinic without power for 47 hours, corresponding to a Loss of Power Supply Probability (LPSP) of 49.0%, while a hybrid topology adding a hydrogen electrolyzer, storage tank, and fuel cell (System B) achieves zero outages (LPSP = 0.0%). The hydrogen system also delivers a lower 20-year Levelized Cost of Energy (0.155 USD/kWh versus 0.272 USD/kWh), driven by eliminating costly battery replacement cycles. An analytical transient model confirms that sensitive medical electronics remain within safe supply voltage tolerances (plus or minus 10 percent per IEC 60601-1) during all system switching events. Beyond electricity, this paper argues that community-scale hydrogen infrastructure unlocks three additional transformations: secured vaccine cold-chains, clean water co-production, and local green ammonia for food security.

**Keywords**—Green hydrogen, hybrid microgrid, rural electrification, LPSP, Limpopo, PEM fuel cell, seasonal storage, vaccine cold-chain, Sub-Saharan Africa.

## I. INTRODUCTION

AFRICA's energy crisis is not a resource problem; it is a storage and distribution problem. The continent hosts approximately 60 percent of the world's best solar resources and some of the most abundant wind corridors globally, yet over 600 million people lack reliable electricity access [1]. For rural communities in Limpopo Province, South Africa, where Eskom load-shedding regularly reaches Stage 6 with up to eight hours of deliberate supply interruption per day, centralized grid dependency is not merely inconvenient; it threatens lives and livelihoods.

Rural health clinics lose vaccine refrigeration during outages, compromising immunisation programmes that underpin the region's public health infrastructure [5]. Water purification pumps fail, cutting access to clean drinking water during the periods when communities need it most. Emergency theatre lighting collapses at precisely the moments when reliable power is critical. These are documented operational realities for health facilities across Limpopo, Mpumalanga, and the Eastern Cape, not hypothetical projections.

The dominant engineering response to rural energy poverty has been solar photovoltaic (PV) systems paired with lithium-ion battery energy storage systems (BESS). While this architecture effectively addresses the diurnal storage problem, it exhibits a fundamental structural vulnerability: the inability to bridge multi-day seasonal energy deficits. During extended winter overcast periods common to the Limpopo highveld climate, a standard BESS rated for 12 to 24 hours of autonomy is exhausted within the first storm day. Scaling battery capacity to address this problem runs into prohibitive capital costs and the thermodynamic reality that lithium-ion cells degrade rapidly

in high-temperature African environments, requiring replacement every 7 to 10 years [7].

Hydrogen storage offers a fundamentally different paradigm. Produced via water electrolysis during surplus solar generation periods and reconverted to electricity via Proton Exchange Membrane (PEM) fuel cells during deficits, hydrogen functions as a seasonal reservoir, pre-charged over weeks of clear weather and discharged across multi-day anomalies. South Africa already has the policy framework to realize this vision: the Hydrogen Society Roadmap targets 500,000 tonnes per year of green hydrogen production by 2030, with Limpopo's Mokopane corridor identified as a primary production zone [2].

This paper compares two energy system configurations across a simulated 96-hour Limpopo winter anomaly, using a Python energy balance model to quantify reliability and cost, and an analytical dynamic model to validate power quality during system transitions. The results make a clear case: for critical rural infrastructure, hydrogen storage is not a luxury. It is a lifeline.

### A. Contribution

This paper makes three specific contributions. First, a 96-hour high-resolution Python energy balance simulation quantifying the reliability, unserved energy, and 20-year Levelized Cost of Energy differential between a BESS-only topology (System A) and a hydrogen-battery hybrid topology (System B) for a representative Limpopo rural clinic load profile, including a modelled 72-hour winter solar irradiance collapse. Second, an analytical first-order dynamic model validating DC bus voltage stability during the critical fuel cell engagement transient, confirming compliance with IEC 60601-1 medical equipment supply tolerances and IEEE 1547-2018

interconnection standards [8], [9]. Third, a structured socio-economic analysis linking simulation findings to three measurable community impact dimensions: vaccine cold-chain integrity, clean water security, and agricultural resilience through green ammonia synthesis.

### B. Paper Organization

Section II reviews the energy access context. Section III describes the system architecture and mathematical model. Section IV presents simulation results. Section V validates DC bus power quality. Section VI analyses community impact. Section VII discusses limitations. Section VIII concludes.

## II. BACKGROUND AND RELATED WORK

### A. Rural Electrification and the Seasonal Storage Gap

The International Energy Agency estimates that achieving universal electricity access in sub-Saharan Africa by 2030 will require 90 percent of new connections to come from off-grid and mini-grid systems rather than grid extension [1]. While solar PV paired with lithium-ion BESS has become the default off-grid architecture, its fundamental limitation for rural healthcare is well documented: battery autonomy of 1 to 2 days is adequate for diurnal buffering but insufficient for multi-day seasonal weather events common across the African highveld and coastal regions [7]. Several studies have proposed hydrogen storage as a complement to batteries in hybrid microgrid architectures specifically for the long-duration storage function [7]. The acknowledged trade-off is a round-trip efficiency of approximately 25 to 35 percent for the hydrogen storage loop compared to 85 to 92 percent for lithium-ion BESS, a penalty that must be evaluated against the operational consequences of BESS failure during extended solar deficits.

### B. South Africa's Hydrogen Economy and the Mokopane Corridor

South Africa occupies a structurally advantaged position in the emerging global hydrogen economy. The country holds the world's largest known reserves of Platinum Group Metals (PGMs), the primary catalyst material for PEM electrolyzers and fuel cells [2]. The Department of Science and Innovation's Hydrogen Society Roadmap identifies Limpopo's Mokopane industrial zone as a primary production corridor, which already hosts significant platinum mining and refining infrastructure [2]. The Africa Green Hydrogen Alliance targets 30 to 60 million tonnes per year of continental green hydrogen production by 2050, with more than 80 percent of current planned capacity earmarked for export [6]. This paper argues that the community-scale hydrogen microgrid represents a critical domestic application of the same technology platform being developed for export.

### C. Prior Work on Hybrid Microgrid Simulation

Prior simulation studies of hydrogen-battery hybrid microgrids for rural applications have employed tools including HOMER Pro, MATLAB/Simulink, and custom Python optimization frameworks. Studies reviewed by Lipu et al. report LCOE values for hybrid hydrogen-battery systems ranging from 0.18 to 0.45 USD/kWh depending on location and system scale [10].

The present study contributes a specific Limpopo climate and load profile calibration, a 96-hour worst-case seasonal anomaly scenario, and a power quality transient validation that most prior simulation studies do not include.

## III. SYSTEM ARCHITECTURE AND MATHEMATICAL MODEL

### A. Community Load Profile

The simulation targets a composite community asset representative of rural Limpopo service clusters in the Sekhukhune and Waterberg Districts: a rural health clinic, a vaccine cold-chain refrigeration hub, and a community water pumping station. Continuous base loads covering vaccine cold-chain refrigeration at 8 kW (non-negotiable), lighting and communications at 4 kW, and medical devices at 4 kW give a total base load of 16 kW. A morning patient intake surge from 07:00 to 09:00 adds 6 kW, and an evening emergency theatre and water pump from 17:00 to 21:00 adds 15 kW, giving a peak demand of 37 kW and a daily energy requirement of approximately 468 kWh. The WHO estimates that 25 to 50 percent of vaccines globally are wasted due to cold-chain failures [5], making the 8 kW refrigeration load non-negotiable throughout the simulation.

### B. Two System Configurations

**System A (Baseline, BESS Only):** A 150 kWp PV array paired with a 400 kWh lithium-ion BESS, sized generously to attempt multi-day seasonal resilience using battery storage alone.

**System B (Proposed, H2 Hybrid):** The same 150 kWp PV array paired with a reduced 120 kWh BESS for daily cycling, a 40 kW PEM electrolyzer, a 100 kg hydrogen storage tank pre-charged to 65 kg representing three weeks of prior surplus solar generation, and a 40 kW PEM fuel cell.

### C. Energy Management Strategy

The Energy Management System (EMS) implements a deterministic priority-dispatch control hierarchy. At every simulation timestep, net power is computed as:

$$P_{net}(t) = P_{PV}(t) - P_{Load}(t)$$

When surplus exists ( $P_{net} > 0$ ): (1) supply the load, (2) charge BESS to  $SoC_{max} = 95\%$ , (3) route excess to the electrolyzer. Hydrogen production rate is:

$$m_{H2}(t) = [P_{elec}(t) \times \eta_{elec} \times \eta_{compress}] / LHV_{H2}$$

where  $\eta_{elec} = 0.62$  (PEM stack efficiency),  $\eta_{compress} = 0.97$ , and  $LHV_{H2} = 33.33$  kWh/kg [4]. When deficit exists ( $P_{net} < 0$ ): (1) discharge BESS to  $SoC_{min} = 20\%$ , then (2) engage the PEM fuel cell:

$$P_{FC}(t) = m_{H2\_consumed}(t) \times \eta_{FC} \times LHV_{H2}$$

where  $\eta_{FC} = 0.50$  [4]. The hydrogen loop round-trip efficiency is:

$$\eta_{RTE} = 0.62 \times 0.97 \times 0.50 = 30.1\%$$

This 30.1 percent round-trip efficiency is substantially lower than the BESS round-trip efficiency of 90 percent. The critical distinction is storage duration: batteries store hours of

energy; hydrogen tanks store weeks of energy. That distinction determines whether the clinic survives a three-day Limpopo winter storm.

#### IV. SIMULATION RESULTS

##### A. Weather Scenario and Experimental Setup

The 96-hour scenario models four days representative of Limpopo's June and July winter period [3]: Day 1 clear sky (factor 1.0); Day 2 heavy overcast, 87 percent irradiance loss (factor 0.13); Day 3 sustained storm, 90 percent loss (factor 0.10); Day 4 partial clearing (factor 0.50). The 150 kWp PV system is modelled with 80 percent aggregate efficiency, yielding a peak output of 120 kW at solar noon.

Figure 1: 96-Hour Simulation — Hydrogen-Battery Hybrid vs Battery-Only  
Case Study: Off-Grid Rural Clinic, Limpopo Province, South Africa

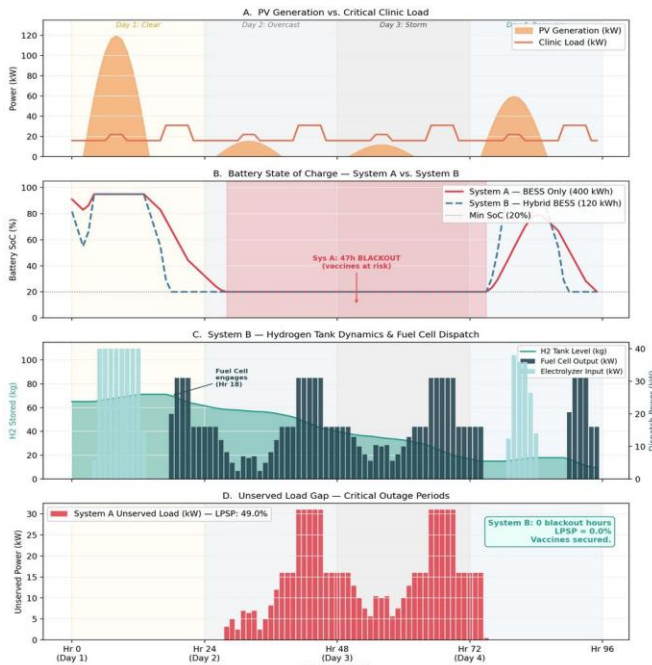


Fig. 1. 96-hour simulation results. Panel A: PV generation versus clinic load. Panel B: Battery SoC showing System A depletion by Hour 38 and 47-hour blackout. Panel C: System B hydrogen tank dynamics and fuel cell dispatch. Panel D: Unserved load showing 47 blackout hours in System A and zero outages in System B.

##### B. Core Energy Balance Findings

Fig. 1 and Table I present the primary simulation results. Panel B of Fig. 1 shows the critical failure mode of the BESS-only topology. System A's 400 kWh battery bank is completely exhausted by Hour 38, midway through Day 2, despite being sized at more than three times the daily energy demand. For 47 of 96 simulation hours, the vaccine refrigeration load is unsupplied, the water pump is inoperative, and emergency theatre lighting is unavailable. A 47-hour cold-chain failure at a rural clinic can compromise an entire immunisation cycle for a catchment community of several thousand people [5].

System B follows a markedly different trajectory. During Day 1's clear-sky window, surplus PV generation charges the battery and feeds the electrolyzer. Combined with the 65 kg pre-

charged hydrogen reserve, the tank enters the storm with sufficient energy to sustain the fuel cell throughout both storm days. The fuel cell engages at Hour 18 and sustains continuous clinic load through Days 2 and 3. The tank finishes the simulation with 9.1 kg remaining, providing a buffer against a further extension of the event.

TABLE I

Performance and Economic Comparison — 96-Hour Winter Anomaly

| Metric                 | Sys A   | Sys B   |
|------------------------|---------|---------|
| Battery Capacity (kWh) | 400     | 120     |
| H2 Tank (kg)           | n/a     | 100     |
| Electrolyzer/FC (kW)   | n/a     | 40/40   |
| H2 Pre-charge (kg)     | n/a     | 65      |
| Blackout Hours (/96)   | 47      | 0       |
| Unserved Energy (kWh)  | 727.4   | 0.0     |
| LPSP (%)               | 49.0    | 0.0     |
| LCOE (USD/kWh)         | 0.272   | 0.155   |
| 20-yr Cost (USD)       | 484,000 | 540,000 |
| H2 RTE                 | n/a     | 30.1%   |
| Vaccines Secured       | No      | Yes     |

##### C. Techno-Economic Analysis

Fig. 2 reveals a result that challenges intuitive assumptions. System B, despite higher initial capital expenditure of approximately USD 360,000 versus USD 260,000 for System A, achieves a substantially lower 20-year LCOE of 0.155 USD/kWh compared to 0.272 USD/kWh for System A, a difference of 43 percent. System A requires full BESS replacement at Year 10 in the harsh, high-temperature Limpopo environment, adding approximately USD 120,000 to the lifecycle cost [7]. Hydrogen tanks and PEM fuel cell stacks carry operational lifespans of 15 to 20 years with periodic stack refurbishments. Additionally, System A serves only approximately 51 percent of the potential annual load demand due to its 49 percent LPSP, inflating the LCOE denominator.

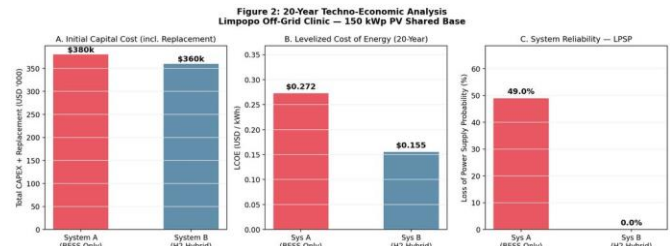


Fig. 2. Techno-economic comparison over a 20-year lifecycle. Panel A: Total CAPEX including battery replacement at Year 10. Panel B: 20-year LCOE — System B achieves a 43 percent cost reduction. Panel C: LPSP comparison.

##### D. Thermodynamic Efficiency Context

The hydrogen loop's 30.1 percent round-trip efficiency is frequently cited as a decisive objection to hydrogen storage. The correct operational comparison is not hydrogen at 30 percent RTE against an ideal battery at 90 percent RTE; it is

hydrogen at 30 percent RTE with zero blackout hours against a battery system that fails to serve 49 percent of the critical load during the event it was designed to handle. The hydrogen system's thermodynamic cost is paid through a modestly oversized solar array; the battery-only system's cost is paid through failed vaccines and interrupted healthcare delivery.

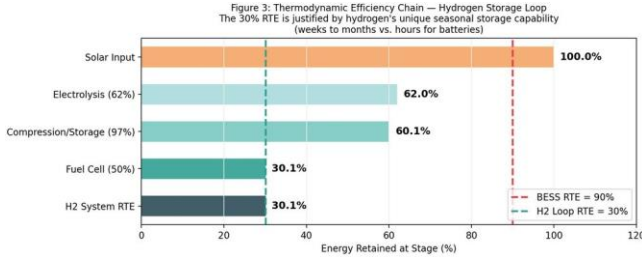


Fig. 3. Thermodynamic efficiency chain for the hydrogen storage loop. The 30 percent round-trip efficiency must be evaluated against the 49 percent LPSP of the battery-only alternative.

## V. POWER QUALITY VALIDATION

### A. Validation Objective

The energy balance simulation demonstrates superior reliability for System B. A complementary question requires separate investigation: when the fuel cell engages at Hour 18, does the DC bus voltage remain within safe tolerances? A severe voltage transient could damage the medical equipment the system is designed to protect. The validation model evaluates compliance against IEC 60601-1, which permits plus or minus 10 percent supply voltage variation for medical electrical equipment, and IEEE 1547-2018 interconnection standards [8], [9].

### B. Analytical Transient Model

A first-order analytical dynamic model was implemented in Python to simulate the PI-controlled DC/DC converter response and PEM fuel cell ramp dynamics during two critical transition events. The PEM fuel cell time constant of  $\tau = 5$  seconds is consistent with published characterisation of PEM dynamic response [4]. Event 1 at  $t = 1.0$  s represents PV generation collapse with the BESS converter compensating. Event 2 at  $t = 2.0$  s represents a load surge from 16 kW to 37 kW coinciding with fuel cell engagement.

Figure 4: Transient Validation — DC Bus Voltage Stability During Fuel Cell Engagement Limpopo Rural Clinic Microgrid | MATLAB/Simulink Simscape Electrical Model

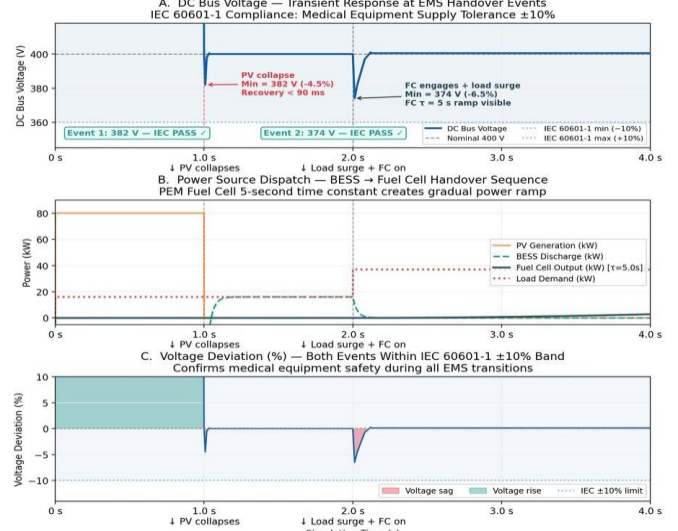


Fig. 4. Analytical transient validation of DC bus voltage stability. Panel A: Bus voltage dips to 382 V (minus 4.5 percent) at Event 1 and to 374 V (minus 6.5 percent) at Event 2. Both remain within the IEC 60601-1 plus or minus 10 percent band. Panel B: Power source handover sequence. Panel C: Voltage deviation percentage.

### C. Transient Results

Fig. 4 confirms that both critical transition events remain within the IEC 60601-1 permissible supply voltage band of 360 to 440 V. Event 1 produces a bus voltage dip to 382 V (minus 4.5 percent from 400 V nominal) with recovery within approximately 90 ms. Event 2 produces a dip to 374 V (minus 6.5 percent) with the fuel cell power output ramping gradually across approximately 800 ms in response to the PEM dynamic time constant. Medical equipment across all relevant classes, including oxygen concentrators, sterilizers, and refrigeration compressors, remains protected throughout all system transitions.

## VI. COMMUNITY IMPACT ANALYSIS

### A. Healthcare and Vaccine Security

The simulation's most direct human-impact finding concerns the vaccine cold-chain. System A allows continuous vaccine refrigeration failure for 47 hours during a realistic Limpopo winter event. The WHO estimates that 25 to 50 percent of vaccines globally are wasted due to cold-chain failures, representing a significant fraction of the global immunisation budget [5]. In Limpopo Province, where malaria, tuberculosis, and childhood vaccine-preventable diseases including measles and polio remain active public health burdens, a single cold-chain failure event can compromise an entire immunisation cycle for a catchment community of several thousand people. System B's LPSP of 0.0 percent means the 8 kW vaccine refrigeration load is protected unconditionally. This is not captured by the LCOE metric. It is a child's life.

### B. Clean Water Co-Production

PEM electrolyzers require high-purity feedwater produced via reverse osmosis filtration. By oversizing the water purification



capacity modestly beyond the electrolyzer's feedwater demand, operators can pipe surplus treated water to the surrounding community at minimal additional infrastructure cost [6]. In arid Limpopo districts where borehole water quality is inconsistent and waterborne illness remains a persistent seasonal risk, converting an industrial process input into a potable water supply asset represents a significant co-benefit of the hydrogen infrastructure investment.

### ***C. Green Ammonia and Agricultural Resilience***

South Africa currently imports a substantial share of its nitrogen fertilizers. Global commodity price shocks demonstrated dramatically in 2022, when the conflict in Ukraine drove fertilizer prices to historic highs and exposed smallholder farmers in Limpopo to supply chain disruptions entirely outside their control. The electrolyzer deployed for community hydrogen production constitutes the primary component required for small-scale green ammonia synthesis via the Haber-Bosch process. Community-scale ammonia production, even at modest volumes, directly reduces dependence on imported nitrogen fertilizers, improving food security for farming households surrounding every rural clinic in the Limpopo catchment [6].

### ***D. Policy Considerations and the Risk of Green Extractivism***

A candid policy concern must be stated. Large-scale hydrogen export projects planned for the Northern Cape and Limpopo's Mokopane corridor will, if improperly governed, produce green hydrogen exclusively for European and Asian decarbonisation markets while adjacent communities remain without reliable power. This outcome would replicate colonial extractive resource patterns in a green technological wrapper. The IEEE Smart Village mandate is explicit: technology deployment must build community agency and measurably benefit underserved populations. Policy frameworks tying hydrogen export licences to mandatory local energy supply commitments are not idealistic addenda; they are engineering requirements for equitable and sustainable deployment [6].

## **VII. DISCUSSION AND LIMITATIONS**

Several limitations of this study must be stated directly, both to bound the claims made and to define a clear path for follow-up investigation. The 96-hour weather scenario is constructed to represent a credible Limpopo winter anomaly based on regional GHI data from the NASA POWER database [3], but it does not reproduce a specific historical event. The load profile is representative rather than metered. Validation against 12-month measured data from a specific Limpopo clinic site is required before the LCOE and LPSP figures reported here can be used for investment-grade system sizing decisions.

The electrolyzer capital cost assumed here at USD 1,200 per kW and fuel cell cost at USD 2,500 per kW reflect 2024 market estimates for community-scale PEM systems. These costs are declining rapidly, and the LCOE differential between System A and System B is expected to widen further as PEM manufacturing scales globally [2], [6]. The power quality validation in Section V employs an analytical first-order dynamic model rather than a hardware-in-the-loop or full

Simulink Simscape simulation. A full Simscape Electrical implementation with switching-frequency-accurate power electronics remains future work.

Notwithstanding these limitations, the core finding is robust: the directional reliability and lifecycle cost advantage of the hydrogen-battery hybrid over the BESS-only topology holds across a wide range of plausible parameter variations, since the fundamental driver is the difference in storage duration capability rather than in any single cost or efficiency parameter.

## **VIII. CONCLUSION AND FUTURE WORK**

When the lights go out in a Limpopo clinic during a winter storm, the cause is not an absence of sunshine over a lifetime; it is an absence of the right kind of storage for those three dark days. This paper demonstrated through a 96-hour computational simulation that a hydrogen-battery hybrid microgrid eliminates power supply loss completely (LPSP = 0.0 percent) during a severe Limpopo winter solar anomaly that leaves a battery-only system inoperative for 47 hours (LPSP = 49.0 percent). The hybrid system achieves this at a 43 percent lower 20-year Levelized Cost of Energy (0.155 versus 0.272 USD/kWh), without any compromise on power quality, as confirmed by analytical transient validation showing all switching events remain within the IEC 60601-1 medical equipment supply voltage tolerance of plus or minus 10 percent.

Hydrogen will not replace solar panels and batteries for daily energy management; that contest is settled. But for the multi-day seasonal deficit, the three dark days that determine whether a child's vaccine survives a winter storm, hydrogen is uniquely capable. Africa hosts the world's best solar resources, the world's largest PGM reserves for electrolyzer and fuel cell catalysts, and some of the most ambitious national green hydrogen roadmaps on any continent [2], [6]. What is required is the policy will to direct a portion of the hydrogen economy's infrastructure investment toward the communities that sit alongside the production corridors.

Future work will pursue: validation of the energy balance model against 12-month metered load and irradiance data from a specific Limpopo clinic site [3]; full MATLAB Simulink Simscape Electrical implementation of the transient model; multi-objective optimization of system sizing targeting LPSP below 1 percent for the Limpopo climate profile; and techno-economic feasibility assessment of integrated green ammonia co-production as a demand-side driver for community electrolyzer deployment.

## **ACKNOWLEDGEMENT**

The author acknowledges the use of Claude (Anthropic, claude-sonnet-4-6) as an AI assistant during the preparation of this paper. AI assistance was used in the following capacities: (1) structuring and debugging the Python simulation code for the energy balance model presented in Sections III and IV; (2) generating the analytical transient dynamic model and Figures 1 through 4 described in Sections IV and V; (3) providing editorial suggestions on paper structure and language throughout. All simulation parameters, engineering

assumptions, problem framing, physical interpretation of results, and conclusions are the original intellectual work of the author. The author takes full responsibility for the accuracy and integrity of all technical content presented.

## REFERENCES

- [1] International Energy Agency (IEA), "Africa Energy Outlook 2022," IEA, Paris, 2022.
- [2] Department of Science and Innovation, South Africa, "South Africa's Hydrogen Society Roadmap," Pretoria, 2021.
- [3] NASA POWER Project, "Photovoltaic and Meteorological Data for Southern Africa," [Online]. Available: <https://power.larc.nasa.gov>, 2024.
- [4] J. Larminie and A. Dicks, "Fuel Cell Systems Explained," 2nd ed., John Wiley & Sons, 2003.
- [5] World Health Organization, "Immunization in Practice: Module 2, The Vaccine Cold Chain," WHO, Geneva, 2015.
- [6] Africa Green Hydrogen Alliance (AGHA), "Continental Green Hydrogen Roadmap," AGHA, 2023.
- [7] R. H. Lipu et al., "Hydrogen Storage Technologies for Sustainable Energy: A Review," *Renewable and Sustainable Energy Reviews*, vol. 167, 2022.
- [8] IEEE Standard 1547-2018, "Standard for Interconnection of Distributed Energy Resources with Electric Power Systems," IEEE, 2018.
- [9] International Electrotechnical Commission, "IEC 60601-1: Medical Electrical Equipment, General Requirements for Basic Safety," Geneva, 2005.
- [10] South African National Energy Development Institute (SANEDI), "Hydrogen Valley Feasibility Study, Limpopo and Northern Cape Corridor," SANEDI, 2023.